Supporting Information

Cobalt Oxide (CoO_x) as an Efficient Hole-extracting Layer for High-performance Inverted Planar Perovskite Solar Cells

Ahmed Esmail Shalan[a], Tomoya Oshikiri[a], Sudhakar Narra[b], Mahmoud M. Elshanawany[b], Kosei Ueno[a], Hui-Ping Wu[b], Keisuke Nakamura[a], Xu Shi[a], Eric Wei-Guang Diau[b] and Hiroaki Misawa[a,b]**

[a] Mr. Ahmed Esmail Shalan, Prof. Dr. K. Ueno, Prof. Dr. T. Oshikiri, Mr. Keisuke Nakamura, Prof. Dr. Xu Shi, Prof. Dr. H. Misawa,

Research Institute for Electronic Science, Hokkaido University, N21, W10, Kita-ku, 001-0021, Sapporo (Japan)

E-mail: misawa@es.hokudai.ac.jp, diau@mail.nctu.edu.tw

[b] Dr. Sudhakar Nara, Mr. Mahmoud M. Elshanawany, Dr. Hui-Ping Wu, Prof. Dr. Eric Wei-Guang Diau, Prof. Dr. H. Misawa,

Department of Applied Chemistry & Institute of Molecular Science, National Chiao Tung University, 1001 Ta Hsueh Road, Hsinchu 30010 (Taiwan)

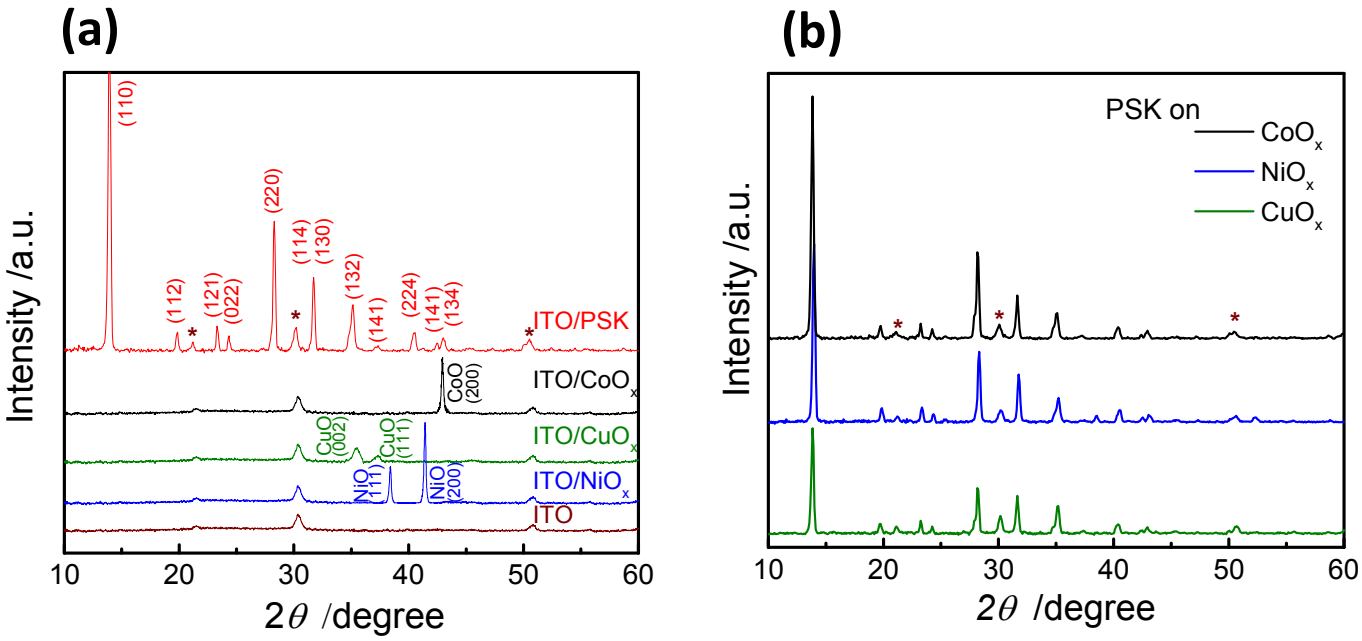

Figure S1. X-ray diffraction (XRD) patterns of (a) [ITO, perovskite (PSK) on ITO, CoO_x on ITO, NiO_x on ITO, CuO_x on ITO], (b) [PSK on CoO_x, NiO_x and CuO_x]. The asterisks in the figures refer to the planes of ITO for all samples. The "*" represents the ITO substrate signals. The assignments of perovskite peaks accorded to XRD reference pattern of perovskite reproduced from the ICSD data base with the code number 250739.

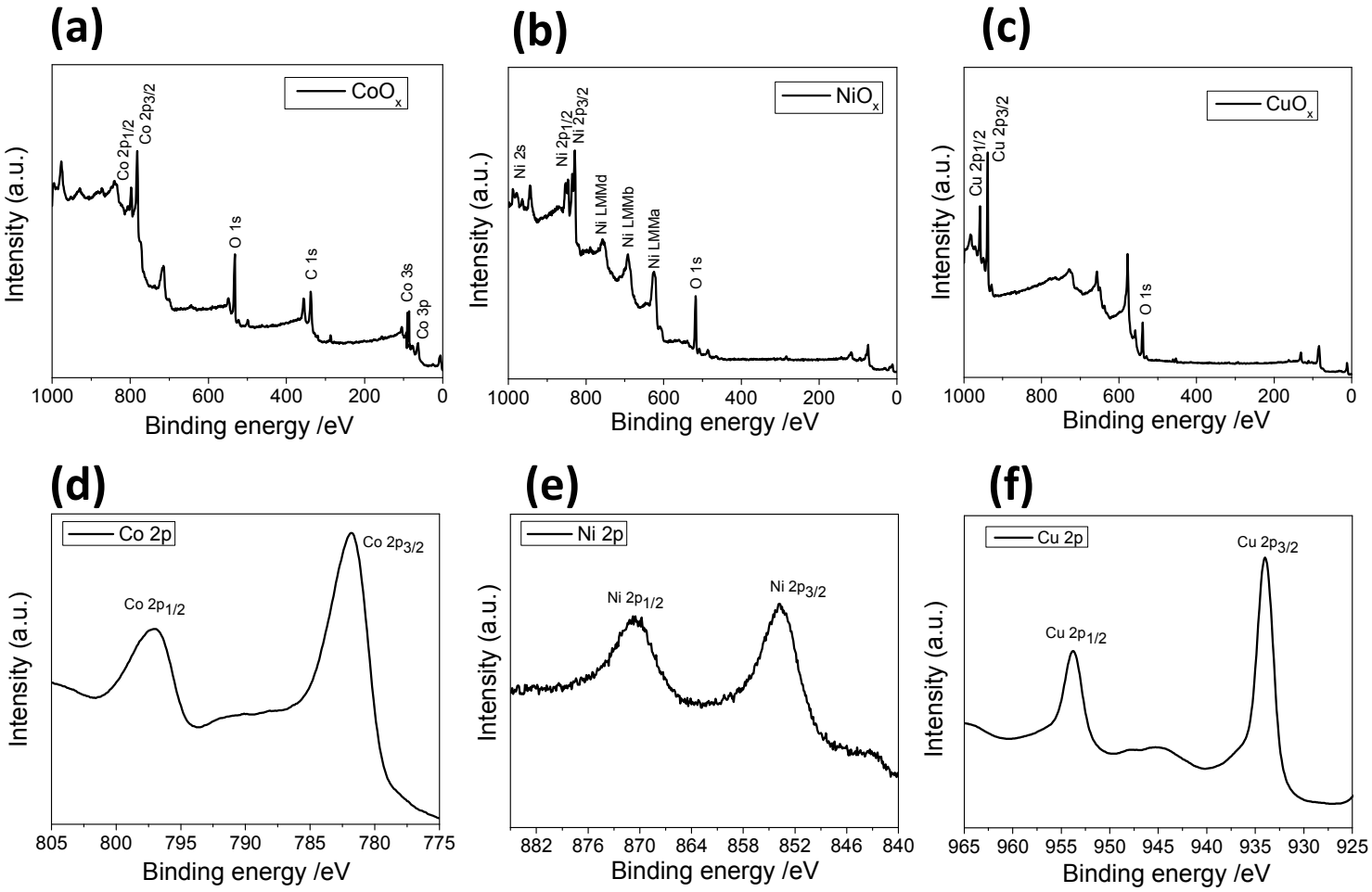

Figure S2. Wide XPS spectra of (a) CoO_x, (b) NiO_x and (c) CuO_x and the corresponding narrow XPS spectra showing Co $2p_{1/2}$ and Co $2p_{3/2}$ for (d) CoO_x, Ni $2p_{1/2}$ and Ni $2p_{3/2}$ for (e) NiO_x and Cu $2p_{1/2}$ and Cu $2p_{3/2}$ for (f) CuO_x. The narrow XPS spectrum of CoO_x(d) exhibits a weak satellite peak between the doublet Co $2p_{3/2}$ and Co $2p_{1/2}$, which is typical evidence for the formation of CoO_x [S1]. The doublets shown in (e) are assigned to the core levels of Ni $2p_{3/2}$ and Ni $2p_{1/2}$, which are comparable with the published results [S2, S3]. The peak position for Ni $2p_{3/2}$ (855.51 ± 0.10 eV) differs from those of metallic Ni (852.7 eV), NiO (853.8 eV) and Ni_2O_3 (856.7 eV) [S4-S6]. The XPS spectrum of CuO_x (f) displays the Cu $2p_{3/2}$ and Cu $2p_{1/2}$ signals located at 934.2 and 954 eV, respectively, which agree satisfactorily with reported values [S7].

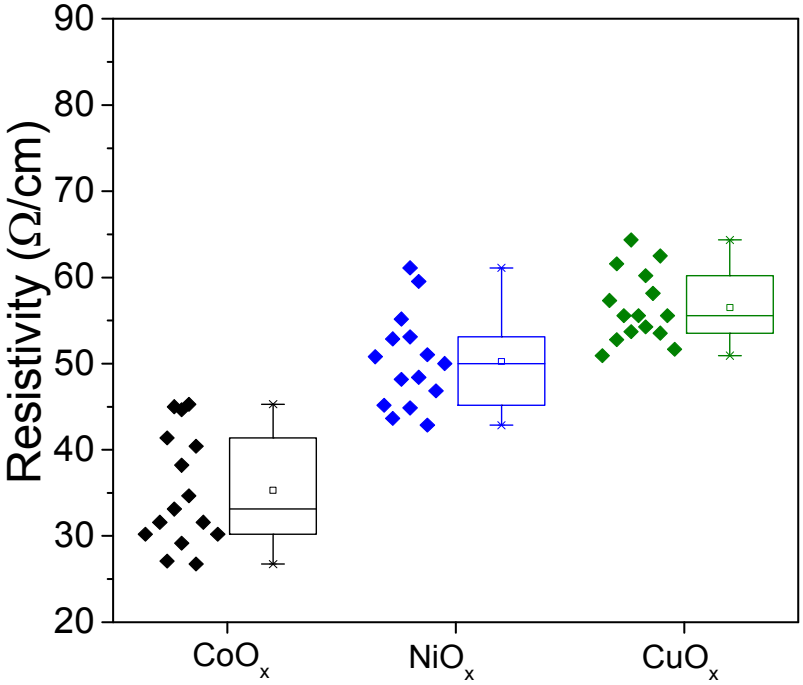

Figure S3. Box chart for the obtained resistivity results calculated from the measurements at 15 different position in the sample surface for the different HELs.

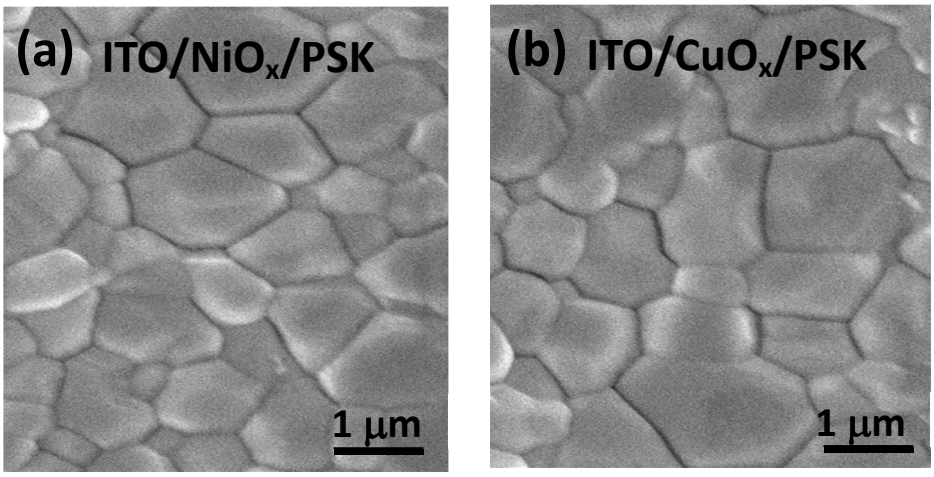

Figure S4. Top-view SEM image of perovskite crystals deposited on (a) NiO_x/ITO and (b) CuO_x/ITO substrates.

Table S1. Photovoltaic parameters of 12 PSC devices made of CoO_x as HEL under simulated AM-1.5G illumination (power density 100 mW cm^{-2}) with active area 0.09 cm^2 for upscaling study of PSCs.

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	16.39	0.994	0.609	9.9
2	16.26	0.989	0.576	9.3
3	16.33	0.988	0.596	9.6
4	16.39	0.987	0.595	9.6
5	16.60	0.985	0.632	10.3
6	16.59	0.984	0.610	9.9
7	16.84	0.982	0.635	10.5
8	16.81	0.983	0.645	10.7
9	16.65	0.984	0.615	10.1
10	16.87	0.982	0.631	10.5
11	16.01	0.981	0.645	10.1
12	16.05	0.981	0.645	10.2
average	16.5±0.3	0.99±0.01	0.62±0.02	10.1±0.4

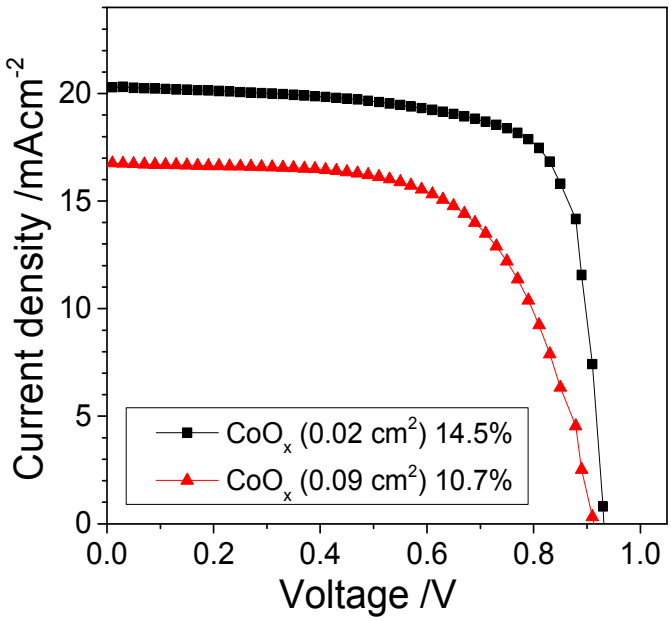

Figure S5. Current-voltage curves of CoO_x, devices as hole-extraction layers (HEL) for upscaling PSCs with the active area (0.09 cm^2) compared to the ones with the active area of (0.02 cm^2).

Table S2. Photovoltaic parameters of 12 PSC devices made of CoO_x as HEL under simulated AM-1.5G illumination (power density 100 mW cm^{-2}) with spin coating parameters of (4000 rpm/30s).

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	19.43	0.930	0.549	9.9
2	19.34	0.960	0.533	9.9
3	19.29	0.922	0.539	9.6
4	19.26	0.930	0.545	9.8
5	19.14	0.973	0.515	9.6
6	19.16	0.958	0.539	9.9
7	19.17	0.963	0.565	10.4
8	19.17	0.978	0.554	10.4
9	19.19	0.974	0.560	10.5
10	18.20	0.973	0.565	10.0
11	18.26	0.969	0.568	10.1
12	18.32	0.968	0.570	10.1
average	19.0±0.5	0.96±0.02	0.55±0.02	10.0±0.3

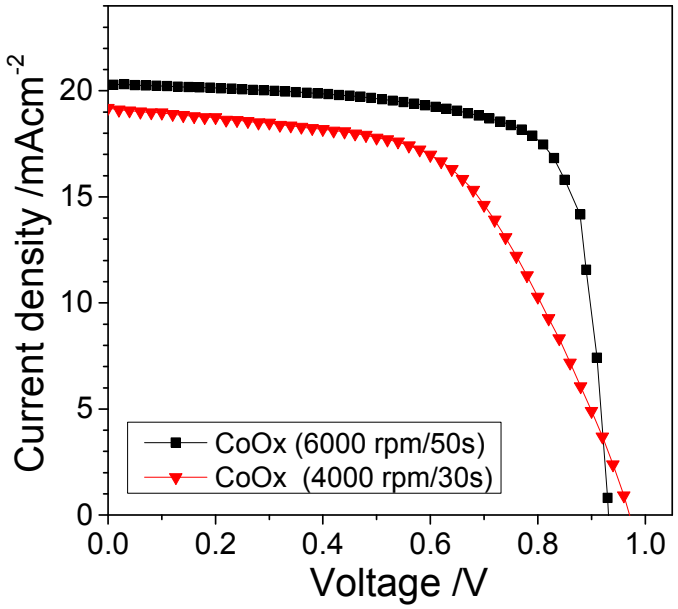

Figure S6. Current-voltage curves of the CoO_x devices as hole-extraction layers (HEL) with spin coating parameters of (6000 rpm/50s) compared to the ones with the spin coating parameters of (4000 rpm/30s).

Table S3. Photovoltaic parameters of 12 PSC devices made of NiO_x as HEL under simulated AM-1.5G illumination (power density 100 mW cm^{-2}) with spin coating parameters of (4000 rpm/30s).

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	19.91	0.960	0.470	8.9
2	20.00	0.957	0.468	8.9
3	20.03	0.949	0.468	8.9
4	19.98	0.959	0.467	8.9
5	20.10	0.944	0.465	8.8
6	20.10	0.944	0.464	8.8
7	19.64	0.954	0.453	8.5
8	19.69	0.952	0.455	8.5
9	19.49	0.938	0.430	7.9
10	19.67	0.950	0.430	8.0
11	19.74	0.950	0.446	8.4
12	19.64	0.951	0.452	8.5
average	19.8±0.2	0.95±0.01	0.46±0.01	8.6±0.4

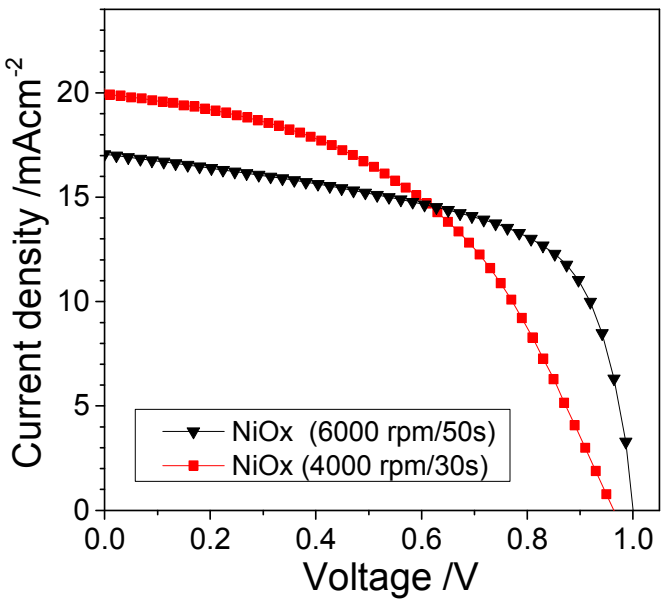

Figure S7. Current-voltage curves of the NiO_x devices as hole-extraction layers (HEL) with spin coating parameters of (6000 rpm/50s) compared to the ones with the spin coating parameters of (4000 rpm/30s).

Table S4. Photovoltaic parameters of 12 PSC devices made of CuO_x as HEL under simulated AM-1.5G illumination (power density 100 mW cm^{-2}) with spin coating parameters of (4000 rpm/30s).

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	18.43	0.905	0.403	6.7
2	18.53	0.919	0.402	6.8
3	18.54	0.933	0.399	6.9
4	18.47	0.919	0.404	6.8
5	18.39	0.901	0.398	6.6
6	18.30	0.935	0.418	7.2
7	16.16	0.937	0.429	6.5
8	15.97	0.942	0.428	6.4
9	16.21	0.946	0.426	6.5
10	16.03	0.946	0.430	6.5
11	16.49	0.931	0.434	6.7
12	16.42	0.936	0.431	6.6

average	17.3±1.2	0.93±0.02	0.42±0.01	6.7±0.2

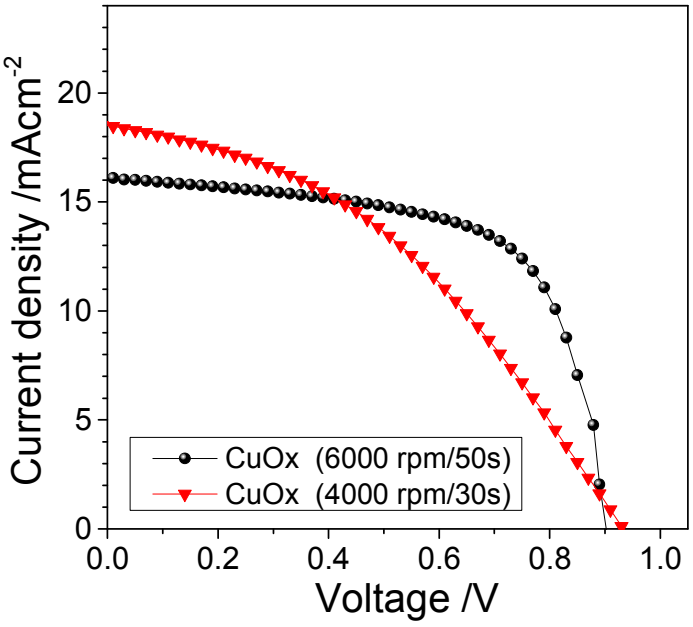

Figure S8. Current-voltage curves of the CuO_x devices as hole-extraction layers (HEL) with spin coating parameters of (6000 rpm/50s) compared to the ones with the spin coating parameters of (4000 rpm/30s).

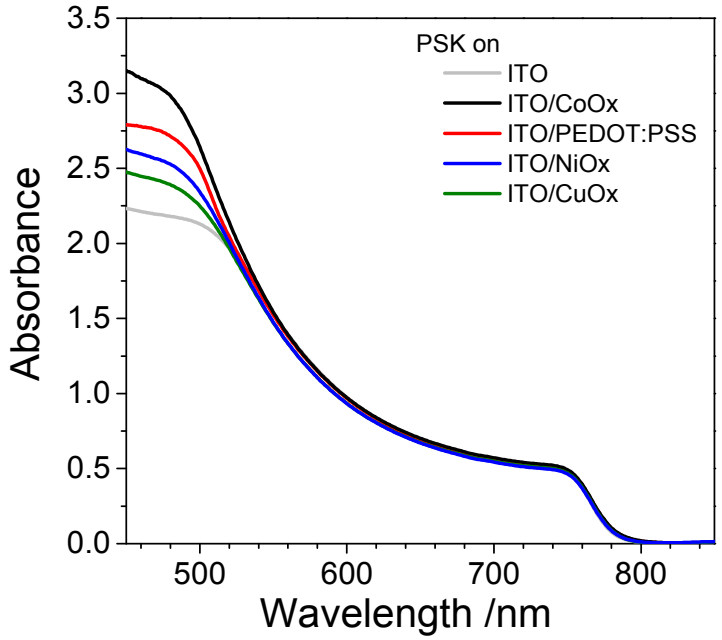

Figure S9. Thin-film absorption spectra of perovskite (PSK) deposited on ITO, ITO/CoO_x, ITO/PEDOT:PSS, ITO/NiO_x and ITO/CuO_x substrates.

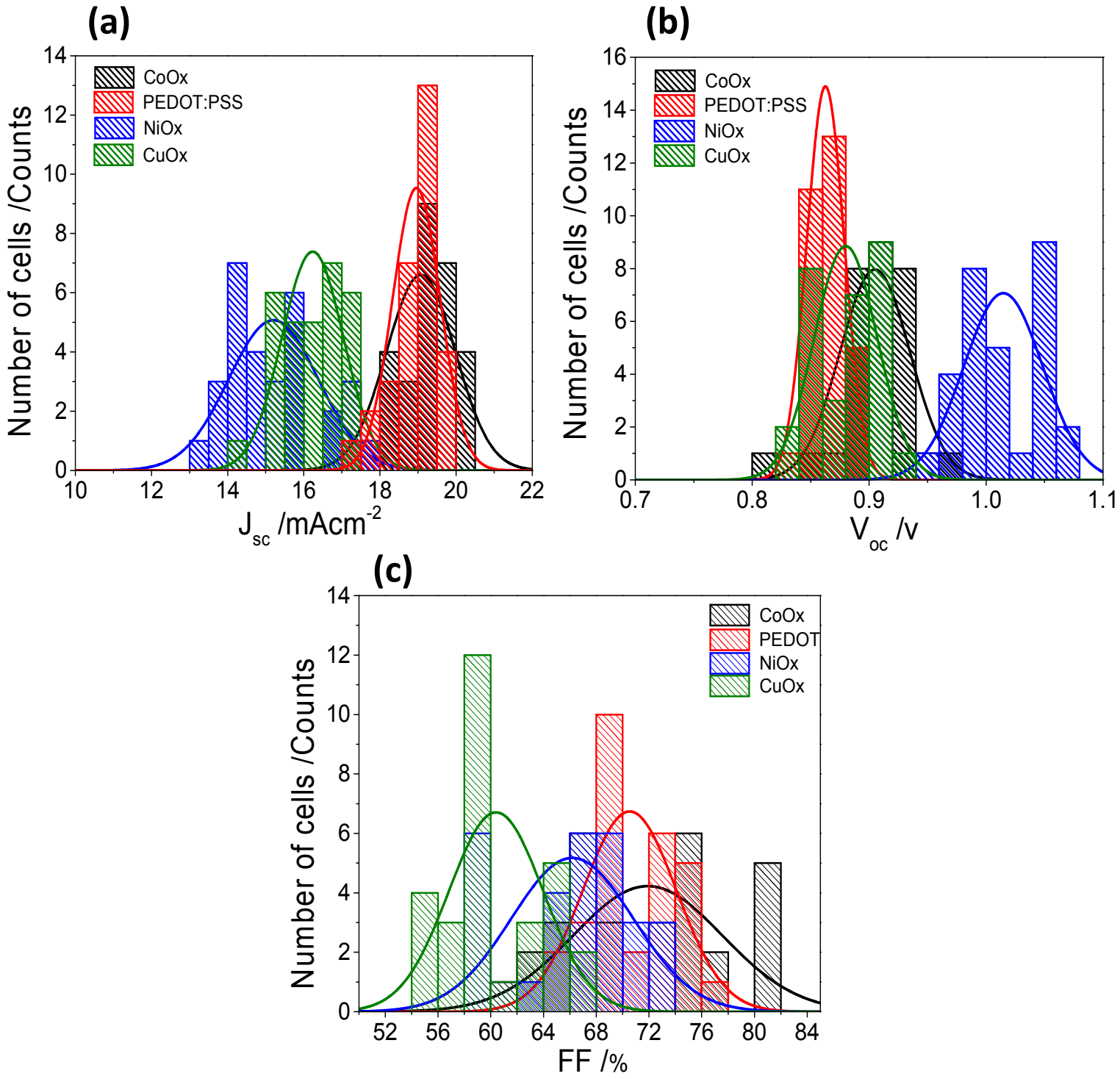

Figure S10. Histograms of the photovoltaic distribution for (a) J_{SC}, (b) V_{OC} and (c) FF for devices made of CoO_x, PEDOT:PSS, NiO_x and CuO_x HEL as indicated.

Table S5. Photovoltaic parameters of 30 PSC devices made of CoO_x as HEL under simulated AM-1.5 G illumination (power density 100 mW cm^{-2}) with active area 0.02 cm^2.

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	19.63	0.876	0.746	12.8
2	19.45	0.920	0.738	13.2
3	19.63	0.932	0.749	13.7
4	18.36	0.927	0.722	12.3
5	19.09	0.908	0.766	13.3
6	18.14	0.907	0.756	12.4
7	19.25	0.889	0.679	11.6
8	19.04	0.888	0.673	11.4
9	19.66	0.932	0.681	12.5
10	18.50	0.947	0.745	13.1
11	18.34	0.967	0.745	13.2
12	19.80	0.911	0.733	13.3
13	18.09	0.906	0.770	12.6
14	17.15	0.895	0.807	12.4
15	17.15	0.899	0.805	12.4
16	17.15	0.899	0.770	11.9
17	18.62	0.802	0.802	11.9
18	18.54	0.846	0.813	12.8
19	19.31	0.921	0.673	11.9
20	19.10	0.931	0.679	12.1
21	19.52	0.919	0.641	11.5
22	19.42	0.922	0.639	11.4
23	19.34	0.923	0.638	11.4
24	20.38	0.900	0.705	12.9
25	20.26	0.907	0.669	12.3
26	20.25	0.886	0.697	12.5
27	20.19	0.890	0.669	12.0
28	19.28	0.891	0.770	13.2
29	19.85	0.910	0.659	11.9
30	19.80	0.907	0.669	12.0
average	19.1±0.90	0.91±0.03	0.72±0.05	12.4±0.7

Table S6. Photovoltaic parameters of 30 PSC devices made of PEDOT:PSS as HEL under simulated AM-1.5 G illumination (power density 100 mW cm^{-2}) with active area 0.02 cm^2.

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	18.51	0.886	0.653	10.7
2	18.34	0.842	0.693	10.7
3	18.41	0.840	0.685	10.6
4	18.28	0.843	0.692	10.7
5	18.67	0.881	0.654	10.8
6	18.57	0.831	0.682	10.5
7	19.17	0.855	0.735	12.0
8	19.25	0.853	0.726	11.9
9	19.36	0.844	0.665	10.9
10	19.29	0.846	0.676	11.0
11	19.24	0.850	0.709	11.6
12	19.21	0.852	0.723	11.8
13	19.62	0.882	0.694	12.0
14	19.09	0.896	0.694	11.8
15	18.96	0.855	0.618	10.0
16	19.38	0.875	0.688	11.7
17	19.27	0.884	0.687	11.7
18	19.21	0.843	0.695	11.3
19	19.96	0.861	0.675	11.6
20	19.35	0.866	0.751	12.6
21	19.94	0.873	0.686	11.9
22	19.30	0.865	0.741	12.3
23	18.91	0.871	0.755	12.4
24	19.13	0.874	0.730	12.2
25	17.83	0.866	0.741	11.4
26	17.19	0.868	0.755	11.3
27	18.79	0.879	0.739	12.2
28	18.78	0.868	0.726	11.8
29	17.94	0.863	0.711	11.0
30	19.83	0.868	0.709	12.2
average	19.0±0.63	0.86±0.02	0.70±0.03	11.5±0.7

Table S7. Photovoltaic parameters of 30 PSC devices made of NiO_x as HEL under simulated AM-1.5G illumination (power density 100 mW cm^{-2}) with active area 0.02 cm^2.

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	14.05	1.061	0.679	10.1
2	14.11	1.052	0.601	8.9
3	14.24	1.062	0.630	9.6
4	13.60	1.050	0.597	8.5
5	13.81	1.057	0.587	8.6
6	14.14	1.051	0.671	9.9
7	14.49	1.016	0.643	9.5
8	15.79	0.986	0.602	9.4
9	13.95	1.012	0.682	9.6
10	14.92	1.022	0.634	9.7
11	14.79	0.996	0.628	9.3
12	14.53	0.978	0.614	8.7
13	14.18	0.997	0.645	9.1
14	13.42	0.981	0.660	8.7
15	15.33	0.990	0.564	8.5
16	15.65	0.995	0.581	9.1
17	15.47	1.010	0.581	9.2
18	14.53	0.975	0.623	8.8
19	14.49	1.016	0.599	8.8
20	15.79	0.985	0.582	9.2
21	15.96	1.012	0.587	9.6
22	15.60	1.050	0.548	8.9
23	15.80	1.057	0.514	8.6
24	16.53	0.975	0.555	8.9
25	17.20	0.968	0.613	10.2
26	17.09	0.945	0.617	9.9
27	17.23	1.041	0.566	10.1
28	17.50	1.055	0.542	10.0
29	16.79	0.996	0.571	9.5
30	15.14	1.051	0.621	9.9
average	15.2±1.18	1.0±0.03	0.61±0.04	9.3±0.5

Table S8. Photovoltaic parameters of 30 PSC devices made of CuO_x as HEL under simulated AM-1.5 G illumination (power density 100 mW cm^{-2}) with active area 0.02 cm^2.

# of cells	J_{sc} /mA cm^{-2}	V_{oc} /V	FF	PCE /%
1	15.79	0.852	0.595	8.0
2	15.66	0.863	0.588	7.9
3	15.34	0.834	0.628	8.1
4	15.18	0.839	0.644	8.2
5	14.25	0.856	0.645	7.9
6	16.23	0.885	0.586	8.4
7	16.98	0.912	0.542	8.3
8	17.31	0.886	0.578	8.9
9	17.32	0.882	0.555	8.5
10	17.21	0.880	0.589	8.9
11	16.63	0.850	0.613	8.7
12	16.52	0.845	0.627	8.8
13	17.45	0.849	0.547	8.1
14	17.39	0.857	0.544	8.1
15	17.29	0.856	0.597	8.8
16	16.35	0.902	0.592	8.7
17	16.19	0.900	0.577	8.4
18	15.57	0.895	0.597	8.3
19	16.52	0.865	0.548	7.7
20	15.34	0.884	0.644	8.7
21	15.17	0.882	0.669	8.9
22	15.45	0.879	0.587	7.9
23	16.82	0.912	0.542	8.3
24	16.60	0.913	0.535	8.1
25	15.36	0.935	0.549	7.9
26	16.52	0.853	0.555	7.8
27	16.44	0.912	0.527	7.9
28	16.28	0.901	0.576	8.4
29	15.95	0.913	0.549	7.9
30	15.91	0.907	0.548	7.9
average	16.2±0.81	0.88±0.03	0.58±0.04	8.3±0.4

References

[S1] Van, E. J.; Wieland, J.L.; Eskes, H.; Kuiper, P.; Sawatzky, G.A.; de Groot, F.M.F.; Turner T.S. Electronic Structure of CoO, Li-doped CoO, and $LiCoO_2$. *Phys. Rev.,* B **1991**; *44*, 6090–6103.

[S2] Pei, G. Q.; Wu, F.; Xia, C.; Zhang, J. G.; Li, X.; Xu, J. Influences of Al Doping Concentration on Structural, Electrical and Optical Properties of $Zn_{0.95}Ni_{0.05}O$ Powders. *Curr. Appl. Phys.* **2008,** *8*, 18–23.

[S3] Wang, H.; Chen, Y.; Wang, H. B.; Zhang, C.; Yang, F. J.; Duan, J. X.; Yang, C. P.; Xu, Y. M.; Zhou, M. J.; Li, Q. High Resolution Transmission Electron Microscopy and Raman Scattering Studies of Room Temperature Ferromagnetic Ni-doped ZnO Nanocrystals. *Appl. Phys. Lett.* **2007**, *90,* 052505.

[S4] Moulder, J. F.; Stickle, W. F.; Sobol, P. E.; Bomben, K. D. Handbook of X-ray Photoelectron Spectroscopy (Eden Prairie, MN USA: Perkin Elmer) 1992, pp 84–85.

[S5] Yu, G. H.; Zeng, L. R.; Zhu, F. W.; Chai, C. L.; Lai, W. Y. Magnetic Properties and X-ray Photoelectron Spectroscopy Study of NiO/NiFe Films Prepared by Magnetron Sputtering. *J. Appl. Phys.* **2001,** *90,* 4039–4043.

[S6] Yin, Z. G.; Chen, N. F.; Yang, F.; Song, S. L.; Chai, C. L.; Zhong, J.; Qian, H. J.; Ibrahim, K. Structural, Magnetic Properties and Photoemission Study of Ni-doped ZnO. *Solid State Commun.,* **2005,** *135,* 430–433.

[S7] Mushtaque, H.; Zafar, H. I.; Mazhar, A. A.; Xianjie, L.; Omer, N.; Magnus, W. Synthesis of Three Dimensional Nickel Cobalt Oxide Nanoneedles on Nickel Foam, Their Characterization and Glucose-sensing Application. *Sensors* **2014,** *14,* 5415–5425.